Syllabus 9758 Statistics Example Questions

Permutations and Combinations

Choosing with "at least" conditions; circular arrangements

Example 1 [TJC 2025 Prelim P2 Q6]

A treasure hunt game is to be played in a 3-storey building. There are 12 specific locations where 5 treasures are to be hidden. Among the 5 treasures, 4 of them are identical. The locations are distributed across the storeys as follows:

- 3 locations on the 1st storey,
- 3 locations on the 2nd storey, and
- 6 locations on the 3rd storey.

Each location can hold at most one treasure.

(a) Find the number of ways to distribute the 5 treasures among the 12 locations for each of the following separate cases:

(i) The 5 treasures can be hidden in any of the 12 locations. [1]

(ii) Exactly 2 treasures must be hidden on the 3rd storey. [2]

(iii) There must be at least one treasure on each of the 1st and 2nd storeys, and at least 2 treasures on the 3rd storey. [3]

Five teams of 2 players each take part in the game. The 10 players and 2 game masters are to be seated around a round table for a briefing.

(b) Find the number of different seating arrangements such that players from the same team must be seated next to each other but the two game masters must not be seated next to each other. [3]

Official solution

(a)(i) Required number of ways $= \binom{12}{5}\binom{5}{1} = 3960$

(a)(ii) Required number of ways $= \binom{6}{2}\binom{6}{3}\binom{5}{1} = 1500$

Alternative method:

Case 1: 2 not identical at 3rd storey $\binom{6}{2} \times 2! \binom{6}{3} = 600$

Case 2: 2 identical at 3rd storey $\binom{6}{2}\binom{6}{3}\frac{3!}{2!}=900$

(a)(iii) Case 1: 2 in 1st storey, 1 in 2nd storey, 2 in 3rd storey

Number of ways $=\binom{3}{2}\binom{3}{1}\binom{6}{2}\binom{5}{1}=675$

Case 2: 1 in 1st storey, 2 in 2nd storey, 2 in 3rd storey

Number of ways $=\binom{3}{1}\binom{3}{2}\binom{6}{2}\binom{5}{1}=675$

Case 3: 1 in 1st storey, 1 in 2nd storey, 3 in 3rd storey

Number of ways $=\binom{3}{1}\binom{3}{1}\binom{6}{3}\binom{5}{1}=900$

Total number of ways $=2250$

(b) Number of ways to arrange the 10 participants $=(5-1)!(2!)^5=768$

Number of ways to slot in the 2 game masters $=\binom{5}{2}2!=20$

Required number $=768\times 20=15360$

Example 2 [ASRJC 2025 Prelim P2 Q11(b)(iii)]

To run an event, the librarian plans to select a team of 12 volunteers from a group of 18. The volunteers comprise three age groups: 5 youths, 6 young adults, and 7 seniors.

(iii) In how many ways can the team be formed if at least one volunteer must be chosen from each age group? [3]

Official solution

(iii) Total number of ways $={}^{18}C_{12}=18564$

No youth $={}^{13}C_{12}=13$

No young adult $={}^{12}C_{12}=1$

Total possible ways $=18564-13-1=18550$

Choosing teams (football)

Example 3 [JPJC 2026 Prelim P2 Q8]

A group of 12 players is available for a street football match. The group consists of 5 attackers, 4 defenders and 3 goalkeepers. A team of 5 players is selected at random from the group.

(a) Find the

(i) number of different teams containing exactly 2 attackers, 2 defenders and 1 goalkeeper, [2]

(ii) probability that the team contains exactly 1 goalkeeper and at most 2 attackers. [3]

The coach has already decided which goalkeeper will play in a match. To help him select from the 5 attackers and 4 defenders, he conducts a passing drill involving these players. For each pass, 1 player is chosen at random to receive the ball. After each pass, the ball is returned to the coach before the next player is chosen. A player may receive the ball more than once. The coach makes 4 passes in this manner. Find the probability that

(b) exactly 3 passes are made to attackers, [2]

(c) either the first 3 passes or the last 3 passes are made to attackers (or both). [3]

Official solution

(a)(i) Number of teams $= {}^5C_2 \times {}^4C_2 \times {}^3C_1 = 180$

(a)(ii)

Case 1: 1 goalkeeper, 2 attackers, 2 defenders $\quad {}^3C_1 \times {}^5C_2 \times {}^4C_2 = 180$

Case 2: 1 goalkeeper, 1 attacker, 3 defenders $\quad {}^3C_1 \times {}^5C_1 \times {}^4C_3 = 60$

Case 3: 1 goalkeeper, 0 attacker, 4 defenders $\quad {}^3C_1 \times {}^5C_0 \times {}^4C_4 = 3$

Total number of ways $= {}^{12}C_5 = 792$

Hence probability that team has at most 2 attackers $= \dfrac{180+60+3}{792} = \dfrac{243}{792}$ or 0.307

(b) Req probability

$= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{AADA}) + \mathrm{P}(\mathrm{ADAA}) + \mathrm{P}(\mathrm{DAAA})$

$= \left(\dfrac{5}{9}\right)^3 \left(\dfrac{4}{9}\right) \times 4 = \dfrac{2000}{6561}$ or 0.305

(c) Method 1:

Req probability $= \mathrm{P}(\mathrm{AAA_}) + \mathrm{P}(\mathrm{_AAA}) - \mathrm{P}(\mathrm{AAAA})$

$= \left(\dfrac{5}{9}\right)^3 + \left(\dfrac{5}{9}\right)^3 - \left(\dfrac{5}{9}\right)^4$

$\dfrac{1625}{6561}$ or 0.248

Method 2:

Req prob $= \mathrm{P}(\mathrm{AAAD}) + \mathrm{P}(\mathrm{DAAA}) + \mathrm{P}(\mathrm{AAAA})$

$= \left(\dfrac{5}{9}\right)^3 \left(\dfrac{4}{9}\right) + \left(\dfrac{4}{9}\right) \left(\dfrac{5}{9}\right)^3 + \left(\dfrac{5}{9}\right)^4$

$\dfrac{1625}{6561}$ or 0.248

Arranging in rows ($2\times n$); round table

Example 4 [ASRJC 2024 Prelim P2 Q8]

An office team of 10 people includes 7 men and 3 women named Anne, Beth, and Cathie. For an upcoming fire drill exercise, 5 individuals will be chosen, each assigned a unique role, to carry out the drill. Determine the number of possible ways to select 5 people from this group of 10

(i) to conduct the fire drill, [1]

(ii) such that at most 1 woman is selected to conduct the fire drill. [2]

(iii) After the fire drill exercise, the 10 people are to hold a discussion at a round table with 10 identical seats. Determine the number of ways in which Beth is seated between Anne and Cathie. [1]

(iv) A group photo of the 10 people, arranged in two rows of five, was taken after the discussion. Determine the number of ways in which Beth is not standing beside Anne or Cathie. [4]

Official solution

(i) Number of ways $= \binom{10}{5}\times 5! = 30240$

(ii) Number of ways $= \left[\binom{7}{5}\binom{3}{0}+\binom{7}{4}\binom{3}{1}\right]5!$

$= (21\times 1+35\times 3)5! = 15120$

(iii) Number of ways $= (8-1)!2! = 10080$

(iv) Case 1 – Beth in 1 row while Anne and Cathie are in another row

$\binom{7}{4}\times 5!\times 5!\times 2 = 1008000$

Case 2 – Beth and one of them in 1 row

$\binom{7}{3}\times 3!\times \binom{2}{1}\times \binom{4}{2}\times 2!\times 5!\times 2 = 1209600$

Case 3 – Anne, Beth and Cathie are in the same row

$\underbrace{\binom{7}{2}\times 2!\times \binom{3}{2}\times 2!\times 2!\times 5!\times 2}_{\text{A and C together}} + \underbrace{\binom{7}{2}\times 2!\times 3!\times 5!\times 2}_{\text{A and C separated}} = 181440$

Thus number of ways $= 1008000+1209600+181440 = 2399040$

Probability

Tree diagram

Example 5 [EJC 2025 Prelim P2 Q6]

An online game consists of three stages. All players start with Stage 1, and no matter whether they win or lose, they proceed to Stage 2. Players are allowed to proceed to Stage 3 only if they have won at least one of the first two stages.

Ming's probability of winning Stage 1 is p.

If Ming wins a stage, his probability of winning the next stage is halved. If he loses a stage, his probability of winning remains unchanged in the next stage.

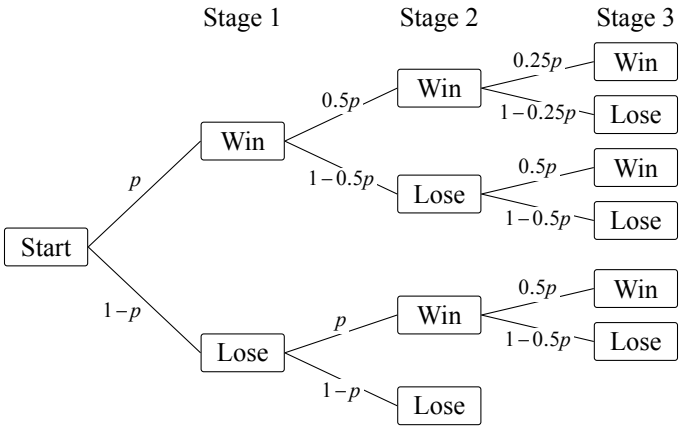
(a) Draw a probability tree diagram to represent all possible outcomes for Ming over the three stages. [3]

(b) Find the range of possible values of p if the probability that Ming wins all three stages is more than 0.1. [2]

(c) It is now known that $p = 0.9$. Given that Ming proceeds to Stage 3, find the probability that he wins exactly two stages out of the three. [4]

Official solution

(a)



(b) $p(0.5p)\ (0.5^2p) > 0.1$

$p^3 > 0.8$

$p > 0.928$

(c) $\mathrm{P}(\text{wins exactly 2 stages out of 3} \mid \text{proceeds to Stage 3})$

$$= \frac{\mathrm{P}(\text{wins exactly 2 stages out of 3} \cap \text{proceeds to Stage 3})}{\mathrm{P}(\text{proceeds to Stage 3})}$$

$$= \frac{0.9(0.45)(1-0.225)+0.9(0.55)(0.45)+0.1(0.9)(0.45)}{1-(0.1)(0.1)}$$

$$= \frac{0.577125}{0.99} = 0.583$$

Venn diagram: range of values of a probability

Example 6 [CJC 2024 Prelim P2 Q10(b)]

The probability of events A, B and C occurring are given by $\mathrm{P}(A) = \frac{8}{25}$, $\mathrm{P}(B) = \frac{61}{100}$ and $\mathrm{P}(C) = \frac{7}{20}$ respectively. It is also known that $\mathrm{P}(A \cap C) = \frac{11}{100}$, $\mathrm{P}(B \mid C) = \frac{13}{35}$ and $\mathrm{P}(A \cap B \cap C) = \frac{7}{100}$.

(i) Find $\mathrm{P}(B \cap C)$. [2]

(ii) If events A and B are independent, find $\mathrm{P}(A \cap B \cap C')$. [2]

(iii) If events A and B are **not** known to be independent, find the greatest and least possible values of $\mathrm{P}(A \cap B \cap C')$. [4]

Official solution

(bi) $\mathrm{P}(B \mid C) = \frac{13}{35}$

$$\frac{\mathrm{P}(B \cap C)}{\mathrm{P}(C)} = \frac{13}{35}$$

$$\frac{\mathrm{P}(B \cap C)}{\frac{7}{20}} = \frac{13}{35}$$

$$\mathrm{P}(B \cap C) = \frac{13}{100}$$

(bii) $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$

$$= \frac{8}{25} \times \frac{61}{100}$$

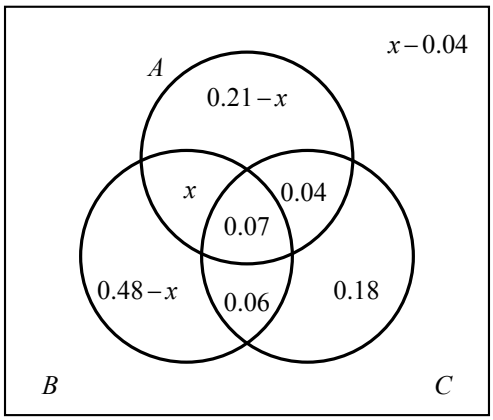
$$= \frac{122}{625}$$

$$\mathrm{P}(A \cap B \cap C')$$

$$= \frac{122}{625} - \frac{7}{100}$$

$$= \frac{313}{2500}$$

(biii)



Let $\mathrm{P}(A\cap B\cap C')=x$ where $0\leqslant x\leqslant 1$ … (1)

Then

$\mathrm{P}(A\cap B'\cap C')=\frac{8}{25}-\frac{11}{100}-x=\frac{21}{100}-x$

$\mathrm{P}(A'\cap B\cap C')=\frac{61}{100}-\frac{13}{100}-x=\frac{12}{25}-x$

$\mathrm{P}(A\cup B\cup C)$

$=\mathrm{P}(C)+\mathrm{P}(A\cap B'\cap C')+\mathrm{P}(A'\cap B\cap C')+\mathrm{P}(A\cap B\cap C')$

$=\frac{7}{20}+\left(\frac{21}{100}-x\right)+\left(\frac{12}{25}-x\right)+x$

$=\frac{26}{25}-x$

Consider the probabilities $\mathrm{P}(A\cap B'\cap C')$, $\mathrm{P}(A'\cap B\cap C')$ and $\mathrm{P}(A\cup B\cup C)$,

$0\leqslant \frac{21}{100}-x\leqslant 1$ and $0\leqslant \frac{12}{25}-x\leqslant 1$ and $0\leqslant \frac{26}{25}-x\leqslant 1$

$-\frac{79}{100}\leqslant x\leqslant \frac{21}{100}$ … (2) $-\frac{13}{25}\leqslant x\leqslant \frac{12}{25}$ … (3) $\frac{1}{25}\leqslant x\leqslant \frac{26}{25}$ … (4)

Taking intersection of inequalities (1) to (4), $\frac{1}{25}\leqslant x\leqslant \frac{21}{100}$

$\min \mathrm{P}(A\cap B\cap C')=\frac{1}{25}$ and $\max \mathrm{P}(A\cap B\cap C')=\frac{21}{100}$

Sum to infinity of a GP

Example 7 [ACJC 2023 Prelim P2 Q9(a)]

A bag contains one green disc, r red discs and $2r$ blue discs, where $r>1$.

(a) Tim and Sim play a game by taking turns to draw a disc from the bag at random. The drawn disc is replaced after each turn. The first person to draw the green disc wins. If Sim started the game first, show that the probability of Sim winning the game is $\frac{3r+1}{6r+1}$. [2]

Official solution

9(a) $\dfrac{1}{3r+1}+\left(\dfrac{3r}{3r+1}\right)^2\times\dfrac{1}{3r+1}+\left(\dfrac{3r}{3r+1}\right)^4\times\dfrac{1}{3r+1}+\ldots$

$=\dfrac{\frac{1}{3r+1}}{1-\left(\frac{3r}{3r+1}\right)^2}=\dfrac{3r+1}{6r+1}$

Contextual probability

Example 8 [HCI 2025 Prelim P2 Q6]

A basketball free throw game involves a team of two players, Ben and John. The game consists of at most three throws and the moment a player makes a successful shot, the team wins, and the game will end.

The game uses the following rules.

- Only one player is selected for each game and the probability that Ben is selected in a game is 0.7.
- The probabilities that Ben and John make a successful shot in any single attempt are 0.1 and 0.07 respectively.
- The shots are independent of each other.

(a) Find the probability that the team wins the game. [3]

(b) The team did not win the game. Find the probability that John was the one who was selected to shoot. [3]

(c) The team attempts the game repeatedly until the first game is won. Find the least number of attempts required such that the probability of winning within n games is at least 0.95. [2]

Official solution

(a) $\mathrm{P}(\text{wins})$

$=\mathrm{P}(\text{wins and Ben selected})+\mathrm{P}(\text{wins and John selected})$

$=\mathrm{P}(\text{wins}\,|\,\text{Ben selected})\mathrm{P}(\text{Ben selected})$

$+\mathrm{P}(\text{wins}\,|\,\text{John selected})\mathrm{P}(\text{John selected})$

$=(0.1+0.9\times0.1+0.9^2\times0.1)\ (0.7)$

$+(0.07+0.93\times0.07+0.93^2\times0.07)\ (0.3)$

$=0.2483929$

OR

$$\begin{aligned}
&\mathrm{P}(\text{wins}) \\
&= \mathrm{P}(\text{wins and Ben selected}) + \mathrm{P}(\text{wins and John selected}) \\
&= \mathrm{P}(\text{wins} \mid \text{Ben selected})\mathrm{P}(\text{Ben selected}) + \\
&\quad \mathrm{P}(\text{wins} \mid \text{John selected})\mathrm{P}(\text{John selected}) \\
&= (1 - 0.9^3)\,(0.7) + (1 - 0.93^3)\,(0.3) \\
&= 0.2483929
\end{aligned}$$

(b)
$$\begin{aligned}
&\mathrm{P}(\text{John selected} \mid \text{did not win}) \\
&= \frac{\mathrm{P}(\text{John selected and did not win})}{\mathrm{P}(\text{did not win})} \\
&= \frac{0.3 \times 0.93^3}{1 - 0.2483929} \\
&= 0.321 \text{ (3sf)}
\end{aligned}$$

(c) Required probability $= 1 -$ P(did not win in all n games)

$$= 1 - (1 - 0.2483929)^n$$

OR

Required probability

$$\begin{aligned}
&= \mathrm{P}(\text{win in 1st game}) + \mathrm{P}(\text{win in 2nd game}) \\
&\quad + \mathrm{P}(\text{win in 3rd game}) + \ldots + \mathrm{P}(\text{win in } n^{\text{th}} \text{ game}) \\
&= 0.2483929 + (1 - 0.2483929)\,(0.2483929) \\
&\quad + (1 - 0.2483929)^2 (0.2483929) + \ldots \\
&\quad + (1 - 0.2483929)^{n-1} (0.2483929) \\
&= \frac{0.2483929\,(1 - (1 - 0.2483929)^n)}{1 - (1 - 0.2483929)} \\
&= 1 - (1 - 0.2483929)^n
\end{aligned}$$

$$1 - (1 - 0.2483929)^n \geqslant 0.95$$

$$(1 - 0.2483929)^n \leqslant 0.05$$

n	$(1 - 0.2483929)^n$
10	$0.0575 > 0.05$
11	$0.0432 < 0.05$

Smallest n is 11.

See also Example 3 (b), (c) for another contextual probability question.

Discrete Random Variables

Unknowns in the table; solving simultaneously

Example 9 [ACJC 2025 Prelim P2 Q8]

The probability distribution of a discrete random variable X is given as follows:

x	-2	-1	0	1	3
$\mathrm{P}(X=x)$	0.1	a	0.3	b	c

Given that $\mathrm{E}(X)=0.9$ and $\mathrm{Var}(X)=2.99$, find the values of a, b and c. [5]

Official solution

$\mathrm{E}(X)=(-2)(\frac{1}{10})+(-1)(a)+(0)(\frac{3}{10})+(1)(b)+(3)(c)$

$$\frac{9}{10}=-\frac{2}{10}-a+b+3c$$

$$-a+b+3c=\frac{11}{10} \quad \ldots\ldots (1)$$

$\mathrm{E}(X^2)=(-2)^2(\frac{1}{10})+(-1)^2(a)+(0)^2(\frac{3}{10})+(1)^2(b)+(3)^2(c)=\frac{4}{10}+a+b+9c$

$\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$

$$\frac{299}{100}=\left(\frac{4}{10}+a+b+9c\right)-\left(\frac{9}{10}\right)^2$$

$$a+b+9c=\frac{34}{10} \quad \ldots\ldots (2)$$

$\sum \mathrm{P}(X=x)=1$

$$\frac{1}{10}+a+\frac{3}{10}+b+c=1$$

$$a+b+c=\frac{6}{10} \quad \ldots\ldots (3)$$

Solving (1), (2) and (3), we have $a=\frac{1}{10}$ or 0.1, $b=\frac{3}{20}$ or 0.15, $c=\frac{7}{20}$ or 0.35

Fair game

Example 10 [ASRJC 2025 Prelim P2 Q7]

Anthony plays a game using a four-sided fair die with the numbers 1, 2, 3 and 4 printed on its sides. Anthony rolls the die and should the outcome be a prime number, he wins an amount equivalent to that number in dollars. Conversely, if the outcome is a non-prime number, Anthony will lose an amount equivalent to that number in dollars. The random variable X represents the amount of money Anthony wins in a roll of the die.

(a) Tabulate the probability distribution of X and find the value of $\mathrm{E}(X)$. [2]

(b) Find the value of the variance of X. [1]

(c) The random variables X_1 and X_2 are two independent observations of X. By drawing a table of outcomes for $|X_1+X_2|$ or otherwise, find $\mathrm{E}(|X_1+X_2|)$. [2]

Assume now that a biased six-sided die with numbers 2, 3, 4, 5, 6 and 7 printed on its sides, is used instead for the same game. It is known that for this biased die, the probability of obtaining a "3" is p, the probability of obtaining a "5" is $2p$ and the remaining four numbers each have an equal probability of occurring.

(d) Find the exact value of p for the game to be fair using the biased die. [3]

Official solution

(a)

x	-4	-1	2	3
$\mathrm{P}(X=x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$\mathrm{E}(X)=\frac{-4}{4}+\frac{-1}{4}+\frac{2}{4}+\frac{3}{4}=0$$

(b) $$\mathrm{Var}(X)=\left[\frac{(-4)^2}{4}+\frac{(-1)^2}{4}+\frac{2^2}{4}+\frac{3^2}{4}\right]-0^2=\frac{15}{2}$$

(c) Let $Y=|X_1+X_2|$

$X_2 \backslash X_1$	-4	-1	2	3
-4	8	5	2	1
-1	5	2	1	2
2	2	1	4	5
3	1	2	5	6

y	1	2	4	5	6	8
$\mathrm{P}(Y=y)$	$\frac{4}{16}$	$\frac{5}{16}$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

$$\mathrm{E}(Y)=\frac{4}{16}+2\left(\frac{5}{16}\right)+4\left(\frac{1}{16}\right)+5\left(\frac{4}{16}\right)+6\left(\frac{1}{16}\right)+8\left(\frac{1}{16}\right)=\frac{13}{4}$$

(d) The new probability distribution for X is as tabulated below:

x	-6	-4	2	3	5	7
$\mathrm{P}(X=x)$	$\frac{1-3p}{4}$	$\frac{1-3p}{4}$	$\frac{1-3p}{4}$	p	$2p$	$\frac{1-3p}{4}$

$$\mathrm{E}(X)=-6\left(\frac{1-3p}{4}\right)-4\left(\frac{1-3p}{4}\right)+2\left(\frac{1-3p}{4}\right)+3p+10p+7\left(\frac{1-3p}{4}\right)=-\left(\frac{1-3p}{4}\right)+13p=-\frac{1}{4}+\frac{55}{4}p$$

For the game to be fair, $-\frac{1}{4}+\frac{55}{4}p=0$

Hence $p=\frac{1}{55}$

Total of two games (not CLT)

Example 11 [CJC 2024 Prelim P2 Q5]

Andrea participates in a game show. In each round of the game, Andrea responds to a series of questions until she answers a question wrongly or she has answered five questions correctly. She is awarded one point for every correct answer and is awarded an additional 3 bonus points if she answers all five questions correctly. It is given that she answers each question independently and the probability she answers each question correctly is $\frac{4}{5}$.

(a) Show that the probability that Andrea obtains 2 points in one round of the game is $\frac{16}{125}$. [1]

(b) Find the probability distribution of the points that Andrea obtains in one round of the game. [2]

(c) Find the expectation and variance of the points that Andrea obtains in one round of the game. [2]

(d) Find the probability that Andrea obtains less than 3 points after two rounds of the game. [3]

Official solution

(a) Probability that Andrea obtains 2 points in a round of the game $= \left(\frac{4}{5}\right)\left(\frac{4}{5}\right)\left(\frac{1}{5}\right) = \frac{16}{125}$

(b) Let X be the random variable denoting the number of points obtained in a round.

x	0	1	2	3	4	8
$P(X=x)$	$\frac{1}{5}$	$\frac{4}{25}$	$\frac{16}{125}$	$\frac{64}{625}$	$\frac{256}{3125}$	$\frac{1024}{3125}$

(c) **Method 1**: Using G.C., Expectation $= 3.67232$ (exact); Variance $= 3.22334^2 = 10.4$ (3 s.f.)

Method 2: $E(X) = 0\left(\frac{1}{5}\right) + 1\left(\frac{4}{25}\right) + 2\left(\frac{16}{125}\right) + 3\left(\frac{64}{625}\right) + 4\left(\frac{256}{3125}\right) + 8\left(\frac{1024}{3125}\right) = 3.67232$

$E(X^2) = 0^2\left(\frac{1}{5}\right) + 1^2\left(\frac{4}{25}\right) + 2^2\left(\frac{16}{125}\right) + 3^2\left(\frac{64}{625}\right) + 4^2\left(\frac{256}{3125}\right) + 8^2\left(\frac{1024}{3125}\right) = 23.87584$

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 23.87584 - (3.67232)^2 = 10.4$ (to 3 s.f.)

(d) Probability that Andrea obtains less than 3 points after two rounds

$= P(X_1 + X_2 < 3) = P(X_1 + X_2 = 0) + P(X_1 + X_2 = 1) + P(X_1 + X_2 = 2)$

$= P(X_1 = 0 \cap X_2 = 0) + 2P(X_1 = 0 \cap X_2 = 1) + 2P(X_1 = 0 \cap X_2 = 2) + P(X_1 = 1 \cap X_2 = 1)$

$= \left(\frac{1}{5}\right)^2 + 2\left(\frac{1}{5}\right)\left(\frac{4}{25}\right) + 2\left(\frac{1}{5}\right)\left(\frac{16}{125}\right) + \left(\frac{4}{25}\right)^2 = \frac{113}{625}$ or 0.1808 (exact)

Expectation of a new variable (new table)

Example 12 [SAJC DRV HW2 Review Q8]

The number of muffins, X, bought by a randomly chosen customer at a muffin shop has probability distribution as shown in the table below:

x	1	2	3	4	5	6	>6
$\mathrm{P}(X=x)$	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	a	$\frac{2}{12}$	$\frac{1}{12}$	0

(i) Find the value of a.

(ii) State the expected value of X.

(iii) Evaluate the variance of X.

Initially, the shopkeeper charges \$2.50 for each muffin, which cost 50 cents to produce.

(iv) Using the answer to part (ii), or otherwise, find the expected profit made per customer.

The shopkeeper decides to promote the sale of muffins by giving a free muffin to all customers who buy 4 or more muffins. As a result, the probability distribution of the number of muffins, Y, taken away by a customer is given in the following table.

y	1	2	3	4	5	6	>6
$\mathrm{P}(Y=y)$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	0	$\frac{20}{36}$	$\frac{6}{36}$	0

(v) Find the expected profit made per customer.

(vi) If the number of customers increases by 8%, is the promotion more profitable for the shopkeeper? Give a reason for your answer.

[Ans: (i) $\frac{1}{4}$ (ii) 3.5 (iii) $1\frac{11}{12}$ (v) \$6.86]

Central Limit Theorem

Example 13 [JPJC 2025 Prelim P2 Q7]

In a game, a player selects one ball at random from a bag which contains four balls numbered '1', '1', '2' and '3'. The player notes the number on the selected ball, and then tosses that number of fair coins. The number of heads obtained is denoted by X and has a probability distribution as follows:

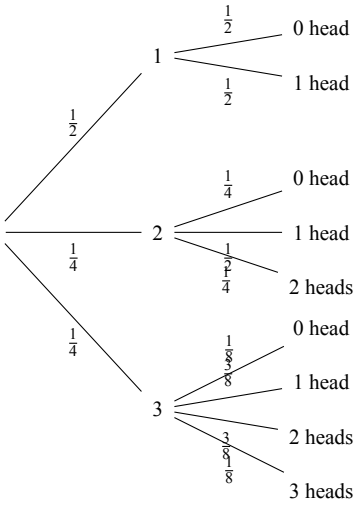
x	0	1	2	3
$\mathrm{P}(X=x)$	a	$\frac{15}{32}$	b	$\frac{1}{32}$

(i) Find the exact values of a and b. [3]

(ii) Find $\mathrm{E}(X)$ and show that $\mathrm{Var}(X)=\frac{39}{64}$. [2]

(iii) The player plays 50 games. Find the probability that the average number of heads obtained per game exceed 1. [2]

Official solution



(i) $a=\mathrm{P}(X=0)=\left(\frac{1}{2}\times\frac{1}{2}\right)+\left(\frac{1}{4}\times\left(\frac{1}{2}\right)^2\right)+\left(\frac{1}{4}\times\left(\frac{1}{2}\right)^3\right)=\frac{11}{32}$

$b=\mathrm{P}(X=2)=\left(\frac{1}{4}\times\left(\frac{1}{2}\right)^2\right)+\left(\frac{1}{4}\times{}^3C_2\times\left(\frac{1}{2}\right)^3\right)=\frac{5}{32}$

Or $b=\mathrm{P}(X=2)=1-\frac{11}{32}-\frac{15}{32}-\frac{1}{32}=\frac{5}{32}$

$a=\frac{11}{32}, b=\frac{5}{32}$

(ii) Probability distribution of X:

x	0	1	2	3
$\mathrm{P}(X=x)$	$\frac{11}{32}$	$\frac{15}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

$\mathrm{E}(X)=\sum x\mathrm{P}(X=x)=(0)\left(\frac{11}{32}\right)+(1)\left(\frac{15}{32}\right)+(2)\left(\frac{5}{32}\right)+(3)\left(\frac{1}{32}\right)=\frac{7}{8}$

$\mathrm{E}(X^2)=\sum x^2\mathrm{P}(X=x)=(0)^2\left(\frac{11}{32}\right)+(1)^2\left(\frac{15}{32}\right)+(2)^2\left(\frac{5}{32}\right)+(3)^2\left(\frac{1}{32}\right)=\frac{11}{8}$

$\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2=\frac{11}{8}-\left(\frac{7}{8}\right)^2=\frac{39}{64}$ (Shown)

(iii) Let X = number of heads obtained in one game; $\overline{X}$ = mean number of heads obtained per game

Since $n=50$ is large, by the Central Limit Theorem, $\overline{X}\sim\mathrm{N}\left(\frac{7}{8},\frac{39/64}{50}\right)=\mathrm{N}\left(\frac{7}{8},\frac{39}{3200}\right)$ approximately

$\mathrm{P}(\overline{X}>1)=0.12876\approx0.129$

Binomial Distribution

Double binomial

Example 14 [ACJC 2025 Prelim P2 Q9]

A fruit seller sources his oranges from a supplier with a large plantation where 1% of the oranges are rotten. The supplier picks the oranges regardless of whether the oranges are rotten or not, and packs 20 oranges in each box at random.

(a) The random variable X denotes the number of rotten oranges in a randomly chosen box. State in context two assumptions for X to follow a binomial distribution. [2]

Assume that X follows a binomial distribution for the rest of this question.

The fruit seller considers a box containing between 2 and 5 rotten oranges inclusive as substandard.

(b) Find the probability that a randomly chosen box is substandard. [1]

(c) Find the expected number of rotten oranges in a box. [1]

A truckload of 10 randomly chosen boxes of oranges is then sent to the fruit seller.

(d) Find the probability that not more than 2 boxes from the entire truckload are substandard. [2]

Upon receiving the oranges, the fruit seller inspects the boxes with these conditions:

- Accept the entire truckload if the first box opened has no rotten oranges.
- Reject the entire truckload if the first box opened has more than 1 rotten orange.
- If the first box has exactly 1 rotten orange, the fruit seller opens a second box.
- If the second box has at most 1 rotten orange, he will accept the entire truckload. Otherwise, he will reject the entire truckload.

(e) Find the probability that he will reject the entire truckload. [2]

Official solution

(a) The probability that an orange is rotten is a constant of 0.01.

Whether an orange is rotten is independent of whether any other oranges are rotten.

(b) Let the random variable X denote the number of rotten oranges in a box of 20 oranges.

$X \sim \mathrm{B}(20, 0.01)$

$\mathrm{P}(2 \leqslant X \leqslant 5) = \mathrm{P}(X \leqslant 5) - \mathrm{P}(X \leqslant 1) = 0.0169$

(c) $\mathrm{E}(X) = (20)(0.01) = 0.2$

(d) Let the random variable Y denote the number of substandard boxes out of the entire truckload of 10 boxes.

$Y \sim \mathrm{B}(10, p)$ where $p = 0.0169$

$\mathrm{P}(Y \leqslant 2) = 0.999$ (3 s.f.)

(e) Let A = {he rejects the entire truckload}

$\mathrm{P}(A) = \mathrm{P}(X \geqslant 2) + \mathrm{P}(X = 1)\mathrm{P}(X \geqslant 2) = [1 - \mathrm{P}(X \leqslant 1)] + \mathrm{P}(X = 1)[1 - \mathrm{P}(X \leqslant 1)] = 0.0196$

The *k*th trial is the *m*th success

Example 15 [YIJC 2024 Prelim P2 Q8]

A manufacturer produces a large number of tumblers of three colours: black, pink and white in the respective ratios $3:2:1$. The tumblers are packed into boxes of 20 each.

Assume that the number of white tumblers in each box follows a binomial distribution.

(a) Find the probability that the number of white tumblers in a randomly chosen box is between 4 and 9 inclusive. [2]

(b) Find the probability that the 19th tumbler chosen from the box is the fourth white tumbler. [2]

It is known that, on average, p % of the tumblers are defective. The number of defective tumblers in a box also follows a binomial distribution. The probability that a box contains at most one defective tumbler is 0.95.

(c) Write down an equation satisfied by p. Hence find the value of p. [3]

Official solution

(a) Let X be the number of white tumblers in a box of 20. $X \sim \mathrm{B}(20, \frac{1}{6})$

$\mathrm{P}(4 \leqslant X \leqslant 9) = \mathrm{P}(X \leqslant 9) - \mathrm{P}(X \leqslant 3) \approx 0.433$

Alternative: $\mathrm{P}(X = 4) + \mathrm{P}(X = 5) + \cdots + \mathrm{P}(X = 9) \approx 0.433$

(b) Let W be the number of white tumblers from 18 tumblers. $W \sim \mathrm{B}(18, \frac{1}{6})$

Required probability $= \mathrm{P}(W = 3) \times \dfrac{1}{6} \approx 0.0409$

(c) Let D be the number of defective tumblers in a box of 20. $D \sim \mathrm{B}(20, 0.01p)$

$\mathrm{P}(D \leqslant 1) = 0.95$

$\mathrm{P}(D=0) + \mathrm{P}(D=1) = 0.95$

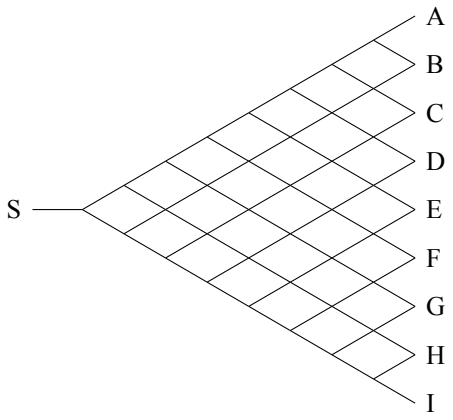
${}^{20}C_0 (0.01p)^0 (1-0.01p)^{20} + {}^{20}C_1 (0.01p)^1 (1-0.01p)^{19} = 0.95$

$(1-0.01p)^{20} + 0.2p(1-0.01p)^{19} = 0.95$

Using G.C., $p \approx 1.81$

Bug turning left and right

Example 16 [N2018 P2 Q6]



In a computer game, a bug moves from left to right through a network of connected paths. The bug starts at S and, at each junction, randomly takes the left fork with probability p or the right fork with probability q, where $q = 1 - p$. The forks taken at each junction are independent. The bug finishes its journey at one of the 9 endpoints labelled A–I (see diagram).

(i) Show that the probability that the bug finishes its journey at D is $56p^5q^3$. [2]

(ii) Given that the probability that the bug finishes its journey at D is greater than the probability that the bug finishes its journey at any one of the other endpoints, find exactly the possible range of values of p. [4]

In another version of the game, the probability that, at each junction, the bug takes the left fork is $0.9p$, the probability that the bug takes the right fork is $0.9q$ and the probability that the bug is swallowed up by a 'black hole' is 0.1.

(iii) Find the probability that, in this version of the game, the bug reaches one of the endpoints A–I, without being swallowed up by a black hole. [1]

[Ans: (ii) $\frac{5}{9} < p < \frac{2}{3}$ (iii) 0.430]

No official SEAB solution is published; answer line as printed with this question in RI 2025 Tutorial S2B Q10.

Range of p given the mode

Example 17 [9758 Specimen Paper (for 2025) P2 Q7]

'Crunchers' are sweets that are sold in packets of 8. Each packet is made up of randomly chosen coloured sweets. On average 15% of Crunchers are yellow.

(a) Find the probability that a randomly chosen packet of Crunchers contains no more than one yellow sweet. [2]

(b) Kev buys 70 randomly chosen packets of Crunchers. Find the probability that at least 50 of these packets contain no more than one yellow sweet. [3]

On average the proportion of Crunchers that are red is p. It is known that the modal number of red sweets in a packet is 3.

(c) Use this information to find exactly the range of values that p can take. [4]

Mark scheme

(a) Use of B(8, 0.15); 0.657

(b) Use of B(70, 0.657); Use of $1-\mathrm{P}(x \leqslant 49)$; 0.189

(c) $\frac{8\times7\times6}{1\times2\times3}p^3(1-p)^5 > \frac{8\times7}{1\times2}p^2(1-p)^6$; $p > \frac{1}{3}$

$\frac{8\times7\times6}{1\times2\times3}p^3(1-p)^5 > \frac{8\times7\times6\times5}{1\times2\times3\times4}p^4(1-p)^4$; $p < \frac{4}{9}$

Central Limit Theorem

Example 18 [TJC 2025 Prelim P2 Q7]

A factory produces porcelain bowls. It is known that, on average, 8% of the bowls are faulty. The bowls are packed in boxes of 20. A box is considered imperfect if it contains more than 2 faulty bowls. Assume that the number of faulty bowls in a box follows a binomial distribution.

(a) Find the probability that a randomly chosen box is imperfect. [2]

These boxes are packed into cartons of 12 boxes each.

(b) Find the probability that, out of 3 randomly chosen cartons, there are 2 cartons that contains fewer than 15 faulty bowls each. [3]

Let Y be the number of imperfect boxes in a randomly chosen carton.

(c) State the values of $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$. [2]

(d) Hence using a suitable approximation, find the probability that the total number of imperfect boxes in a random sample of 35 cartons is at most 85. [2]

Official solution

(a) Let X be the number of faulty bowls in a box of 20 bowls. $X \sim \mathrm{B}(20, 0.08)$

$\mathrm{P}(X > 2) = 1 - \mathrm{P}(X \leqslant 2) = 1 - 0.78795 = 0.21205$ (5 s.f.) $= 0.212$ (3 s.f.)

(b) Let W be the number of faulty bowls in a carton (of 12 boxes). $W \sim \mathrm{B}(12 \times 20, 0.08)$, i.e. $W \sim \mathrm{B}(240, 0.08)$

$\mathrm{P}(W < 15) = \mathrm{P}(W \leqslant 14) = 0.12933$ (5 s.f.)

Required Probability $= {}^{3}C_{2}(0.12933)^{2}(1 - 0.12933) = 0.0437$ (3 s.f.)

OR Let A be the number of cartons out of 3 contains fewer than 15 faulty bowls each. $A \sim \mathrm{B}(3, 0.12933)$

Required Probability $= \mathrm{P}(A = 2) = 0.0437$ (3 s.f.)

(c) $Y \sim \mathrm{B}(12, 0.21205)$

$\mathrm{E}(Y) = np = 12 \times 0.21205 = 2.5446 = 2.54$ (3 s.f.)

$\mathrm{Var}(Y) = np(1-p) = 12 \times 0.21205 \times (1 - 0.21205) = 2.0050$ (5 s.f.) $= 2.01$ (3s.f.)

(d) $Y \sim \mathrm{B}(12, 0.21205)$. Let $T = Y_1 + \cdots + Y_{35}$.

Since sample size 35 is large, by Central Limit Theorem,

$T \sim \mathrm{N}(35 \times 2.5446,\ 35 \times 2.0050)$ approximately

i.e. $T \sim \mathrm{N}(89.061,\ 70.175)$ approximately

Required probability $= \mathrm{P}(T \leqslant 85) = 0.314$ (3 s.f.)